

CSCE 46103/56103 Artificial Intelligence – Fall 2026

Expected Utility and Value of Information

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Today's Topics

1. **Expected Utility:** preferences, action selection, and utility estimates
2. **Value of Information:** observations, improved decisions, and information costs
3. **MDP Formulation:** states, actions, transition probabilities, and rewards
4. **Policies and Returns:** sequential decisions, horizons, and discounted utility

A Television Game Show

You have reached the final choice in a game show.

Keep the Prize

Take \$1,000,000 for certain.

Accept the Gamble

Flip a fair coin:

- ▶ Heads: receive \$0.
- ▶ Tails: receive \$2,500,000.

Which would you choose? What information about your preferences would help explain that choice?

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The Decision Question

A larger expected monetary payoff can have a smaller expected utility.

Actions and Outcomes

Let k be the contestant's current wealth, before receiving a prize.

Decision	Outcome Wealth	Probability
Keep the prize	$k + 1,000,000$	1
Accept the gamble	k	1/2
	$k + 2,500,000$	1/2

- ▶ A decision d selects an action. An outcome o describes what happens after it.
- ▶ $U(o)$ measures how the contestant values that outcome.
- ▶ Winning no prize leaves wealth at k ; it does not remove existing wealth.

Expected Utility

For a decision d and available information e , average the utilities of its possible outcomes:

$$EU(d \mid e) = \sum_o P(o \mid d, e)U(o).$$

- ▶ Probabilities describe how likely the outcomes are under the chosen action.
- ▶ Utilities describe how desirable the outcomes are.
- ▶ The outcome probabilities for each action sum to one.

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Maximum Expected Utility

$$d^*(e) \in \arg \max_d EU(d \mid e).$$

Choose an action with the largest expected utility under the model.

Expected Monetary Value

First compare the expected prize amounts:

$$\text{EMV}(\text{keep}) = \$1,000,000,$$

$$\begin{aligned}\text{EMV}(\text{gamble}) &= \frac{1}{2}(\$0) + \frac{1}{2}(\$2,500,000) \\ &= \$1,250,000.\end{aligned}$$

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A Higher Expected Prize

The gamble has a larger expected monetary value. To compare expected utilities, we also need the contestant's preferences over the possible outcomes.

Utility Assignments

Consider a contestant with the following utilities:

Outcome	Total Wealth	Utility
No prize	k	5
Guaranteed prize	$k + 1,000,000$	8
Larger prize	$k + 2,500,000$	9

- ▶ More money has greater utility for this contestant.
- ▶ The utility gain from the first million is $8 - 5 = 3$.
- ▶ The additional 1.5 million adds only $9 - 8 = 1$ utility unit.

The Preferred Game-Show Decision

Using these utility assignments,

$$\text{EU}(\text{keep}) = 8,$$

$$\text{EU}(\text{gamble}) = \frac{1}{2}(5) + \frac{1}{2}(9) = 7.$$

The Preferred Game-Show Decision

Using these utility assignments,

$$EU(\text{keep}) = 8,$$

$$EU(\text{gamble}) = \frac{1}{2}(5) + \frac{1}{2}(9) = 7.$$

Choice

Keep the guaranteed prize, because $8 > 7$.

The gamble has the larger expected monetary value, while keeping the prize has the larger expected utility for these preferences.

Money, Utility, and Risk

The Game-Show Contestant

The sure million delivers most of the utility gain offered by the larger prize.

Giving up that certainty for a chance at more money lowers expected utility.

The probabilities can be identical while different utility functions justify different actions.

A Risk-Neutral Agent

Utility is linear in wealth over the relevant range.

Maximizing expected utility then agrees with maximizing expected money, so this agent accepts the gamble.

Utility Scales

A positive affine transformation preserves the preferred action:

$$U'(o) = \alpha U(o) + \beta, \quad \alpha > 0.$$

For the game show, use $U' = (U - 5)/4$:

	No Prize	Sure Prize	Larger Prize
U	5	8	9
U'	0	0.75	1

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	No Prize	Sure Prize	Larger Prize
U	5	8	9
U'	0	0.75	1

The transformed expected utilities are 0.75 for keeping and 0.50 for gambling. The choice stays the same.

Checkpoint: A Different Coin

Keep the same prizes and utilities, but change the chance of winning the larger prize.

Modified Probability

The coin gives the larger prize with probability $q = 0.8$ and no prize with probability $1 - q = 0.2$.

1. What is the expected utility of the gamble?
2. Should the contestant gamble or take the guaranteed prize?
3. At what value of q do the two choices have equal expected utility?

Checkpoint Solution

For any probability q of the larger prize,

$$\text{EU}(\text{gamble}) = (1 - q)5 + q9 = 5 + 4q.$$

At $q = 0.8$,

$$\text{EU}(\text{gamble}) = 8.2 > 8 = \text{EU}(\text{keep}).$$

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Indifference Threshold

$$5 + 4q = 8 \iff q = 0.75.$$

Gamble when $q > 0.75$, keep the prize when $q < 0.75$, and either action is optimal at equality.

Expected Utility and Realized Utility

In the checkpoint with $q = 0.8$, gambling has expected utility 8.2.

Realized Outcome	Probability	Utility
No prize	0.2	5
Larger prize	0.8	9

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- ▶ The contestant receives utility 5 or 9 in a single play.
- ▶ The average tends to 8.2 over independent repetitions of this same decision problem.
- ▶ An unlucky outcome is possible even when the probabilities and expected utilities are correct.

Estimated Expected Utilities

In complex problems, the agent often computes an estimate of expected utility:

$$\widehat{EU}(d) = EU(d) + \varepsilon_d.$$

- ▶ The model can omit relevant details.
- ▶ Computing the exact expectation can be too expensive.
- ▶ Even an unbiased estimate can be too high or too low in a particular instance.

Action Selection

$$\hat{d} \in \arg \max_d \widehat{EU}(d).$$

We select using estimates, while outcomes follow the underlying problem.

An Estimation Model

There are k available choices. Each has true expected utility 0:

$$\text{EU}(d_i) = 0, \quad \widehat{\text{EU}}(d_i) = \varepsilon_i, \quad i = 1, \dots, k.$$

The estimation errors are independent standard normal variables:

$$\varepsilon_i \sim \mathcal{N}(0, 1).$$

- ▶ Each fixed choice has an unbiased estimate: $\mathbb{E}[\varepsilon_i] = 0$.
- ▶ We choose the option with the largest observed estimate.
- ▶ The true values are given for our analysis. The selecting agent only sees the estimates.

Selection of an Optimistic Estimate

One illustrative set of estimates under this model is:

Choice	True Expected Utility	Estimate
d_1	0	-0.4
d_2	0	1.1
d_3	0	0.2

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Choice	True Expected Utility	Estimate
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d_2	0	1.1
d_3	0	0.2

Selected Choice

We select d_2 because its estimate is 1.1. Its true expected utility is still 0.

Maximizing the estimates favors choices whose errors happen to be positive.

The Optimizer's Curse

For the selected action $\hat{d}$, the reported estimate is

$$\widehat{\text{EU}}(\hat{d}) = \max_i \varepsilon_i, \quad \text{EU}(\hat{d}) = 0.$$

Number of Choices	Mean Selected Estimate	True Expected Utility
1	0	0
3	About 0.85	0
30	About 2	0

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Averaged over repeated estimation, the selected estimate is too optimistic. Individual estimates can be unbiased while selection introduces bias.

Bayesian Adjustment of Utility Estimates

Treat an estimated utility as evidence about an unknown true expected utility.

An Explicit Error Model

Let u denote a possible true expected utility and $\hat{u}$ its noisy estimate:

$$p(u \mid \hat{u}) \propto p(\hat{u} \mid u)p(u).$$

Here p denotes a probability density.

- ▶ The prior expresses plausible utility values before seeing the estimate.
- ▶ The likelihood describes the estimation error.
- ▶ Compare posterior expected utilities, accounting for the available estimates.

Checkpoint: Two Sources of Disappointment

1. In the game-show variation, the larger prize has probability 0.8. The contestant chooses the gamble but receives no prize.
2. In the estimation model, the agent selects the highest noisy estimate and finds that the selected action's true expected utility is lower.

Which case concerns randomness in an outcome, and which concerns selection among utility estimates?

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Which case concerns randomness in an outcome, and which concerns selection among utility estimates?

Distinction

The first can occur with an exact model. The second illustrates the optimizer's curse: selection favors optimistic estimation errors.

Oil-Drilling Rights

A company can buy one of n blocks. Exactly one contains oil, and all blocks are equally likely.

A Numerical Instance

Use $n = 3$ blocks and oil worth $C = \$12$ million. Rights to each block cost $C/n = \$4$ million.

- ▶ Buy at most one block, or buy none.
- ▶ The company is risk-neutral, so it maximizes expected monetary profit.
- ▶ Oil value excludes the rights price and any survey fee in our accounting.
- ▶ A private survey arrives before purchase. Rights prices remain unchanged.

Buying Without a Survey

Each oil location has probability $1/3$. Payoffs below are profits in millions of dollars after buying the rights.

Decision	Oil in Block 1	Oil in Block 2	Oil in Block 3
Buy block 1	8	-4	-4
Buy block 2	-4	8	-4
Buy block 3	-4	-4	8
Buy none	0	0	0

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Buy none	0	0	0

Buying any block has expected profit

$$\frac{1}{3}(8) + \frac{2}{3}(-4) = 0.$$

Buying none also gives 0. The current best expected profit is $V = 0$.

A Perfect Survey of Block 3

The survey reports whether block 3 contains oil.

Oil in Block 3

This result has probability $1/3$.

$$P(\text{oil in 3} \mid \text{yes}) = 1.$$

Buy block 3. Profit is

$$12 - 4 = 8.$$

No Oil in Block 3

This result has probability $2/3$.

$$P(\text{oil in 1} \mid \text{no}) = \frac{1}{2}.$$

Blocks 1 and 2 are equally likely. Buy either one. Expected profit is

$$\frac{1}{2}(12) - 4 = 2.$$

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An empty survey leaves uncertainty about which of the other two blocks contains oil.

Expected Value After the Survey

Before seeing the result, average the value of the best action in each branch:

$$V_{\text{survey}} = \frac{1}{3}(8) + \frac{2}{3}(2) = 4.$$

Value of the Information

$$\text{VPI}(\text{Oil}_3) = V_{\text{survey}} - V = 4 - 0 = 4.$$

The survey of block 3 is worth \$4 million before its fee.

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The survey of block 3 is worth \$4 million before its fee.

The company can choose block 3 after a positive result and a different block after a negative result. Its action depends on the observation.

A General Oil Model

Let $n \geq 2$. Exactly one block contains oil worth C , and each block's rights cost C/n .

Survey Result	Probability	Best Purchase	Expected Profit
Oil in surveyed block	$1/n$	That block	$C - C/n$
No oil there	$(n - 1)/n$	Another block	$C/(n - 1) - C/n$

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$$V_{\text{survey}} = \frac{1}{n} \left(C - \frac{C}{n} \right) + \frac{n - 1}{n} \left(\frac{C}{n - 1} - \frac{C}{n} \right) = \frac{C}{n}.$$

The baseline is 0, so the survey's value is C/n .

Value of Information: General Formula

Let X be an observable variable and e the information already available.

$$V(e) = \max_d \text{EU}(d \mid e).$$

After observing $X = x$, choose the best action for that result:

$$V_X(e) = \sum_x P(x \mid e) \max_d \text{EU}(d \mid e, x).$$

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Expected Improvement

$$\text{VPI}(X \mid e) = V_X(e) - V(e).$$

Here “perfect” means that we learn the value of X exactly.

Action Selection After Observation

Observation Before the Purchase

$$\sum_x P(x \mid e) \max_d EU(d \mid e, x)$$

The chosen block can depend on the survey result.

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Purchase Committed in Advance

$$\max_d \sum_x P(x \mid e) \text{EU}(d \mid e, x) = \max_d \text{EU}(d \mid e)$$

One purchase must apply to every result. In the oil example, that gives expected profit 0.

The survey changes beliefs, while the oil location and available purchases remain the same.

Information About One Variable

Status of Block 3

The survey reveals whether block 3 contains oil. If it is empty, blocks 1 and 2 remain equally likely.

$$\text{VPI}(\text{Oil}_3) = 4.$$

The Entire Oil Location

If we learn exactly which block contains oil, we can always buy that block.

$$V_{\text{location}} = 12 - 4 = 8.$$

Specify which variable the information reveals. Perfect information about one variable can leave other uncertainty unresolved.

Exact Observations and Noisy Sensors

A measured reading differs from the underlying quantity:

$$\text{Thermometer} = 37$$

can be known exactly while the true temperature remains uncertain.

Measurement Model

$$P(\text{Thermometer} \mid \text{Temperature})$$

The sensor model represents measurement error. Observing the reading updates the distribution over true temperature.

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The same information-value formula can evaluate a noisy test: average over its possible readings, updating beliefs and choosing an action after each reading.

Information Costs

For a risk-neutral company, a survey fee c reduces expected monetary profit by c .

$$V_{\text{buy survey}}(e) = V_X(e) - c.$$

Buying the survey is strictly preferable to acting immediately when

$$\boxed{\text{VPI}(X \mid e) > c.}$$

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Compatible Units

In the oil example, compare monetary profit with a monetary fee. For general utilities, include the information cost in the outcome utility, or express it as an additive cost on that utility scale.

Nonnegative Value of Free Information

Let d_0 be an optimal action before the new observation. For every possible result x ,

$$\max_d \text{EU}(d \mid e, x) \geq \text{EU}(d_0 \mid e, x).$$

Averaging over x gives

$$V_X(e) \geq \sum_x P(x \mid e) \text{EU}(d_0 \mid e, x) = V(e).$$

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Assumptions

The agent may ignore the information. Observation has no cost or physical effect, and the same actions remain available.

Thus $\text{VPI}(X \mid e) \geq 0$, as an expectation before receiving the result.

Mountain Routes: A Clear Preference

Consider two winter routes through a mountain range:

Route 1

A straight highway through a tunnel,
unlikely to be blocked.

Route 2

A winding dirt road over the mountain,
possibly blocked by snow.

A satellite report could provide more information about current road conditions.

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Information Value

In this example, the highway is very likely to remain preferable. Reports rarely change the preferred route, so their expected value is small.

Mountain Routes: Consequential Uncertainty

Now the choice is between two winding dirt roads in winter, and the vehicle carries a seriously injured passenger.

- ▶ Either road may be blocked by snow.
- ▶ Their current expected utilities can be close.
- ▶ Choosing an open road instead of a blocked one can make a large difference.

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Information Value

A satellite report can change the preferred route and substantially improve the outcome. Obtaining it can have high expected value.

Mountain Routes: Small Differences

Consider the same two dirt roads in summer, when snow blockage is unlikely.

- ▶ A report might reveal a more scenic route.
- ▶ It might also reveal a wet road after recent rain.
- ▶ This information can change the route choice, while making only a small difference in utility.

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Consider the same two dirt roads in summer, when snow blockage is unlikely.

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Information Value

Information can change the preferred action and still have little value. Both the chance of changing the plan and the size of the improvement matter.

Zero Value and Repeated Information

The Same Action After Every Result

If one action remains optimal for every possible observation, updating beliefs leaves the expected value of the best decision unchanged:

$$\text{VPI}(X \mid e) = 0.$$

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A Result Already Known

After learning the exact survey result, receiving the identical result again adds no information. The second copy has zero additional value.

The value of information depends on the evidence already available. Values of different observations do not generally add independently.

Checkpoint: The Price of a Survey

Return to three oil blocks, oil value \$12 million, and a rights price of \$4 million per block.

Survey of Block 3

Before its fee, the survey raises the best expected profit from \$0 to \$4 million.

1. Should the company buy this survey for \$3 million?
2. Should it buy the same survey for \$5 million?
3. If the survey reports an empty block, does the company know which remaining block contains oil?

Checkpoint Solution

All values below are in millions of dollars:

Option	Expected Profit After Fee	Preference
Act immediately	0	Baseline
Survey at fee 3	$4 - 3 = 1$	Buy the survey
Survey at fee 5	$4 - 5 = -1$	Act immediately

Checkpoint Solution

All values below are in millions of dollars:

Option	Expected Profit After Fee	Preference
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- ▶ A fee of 4 makes the company indifferent between surveying and acting immediately.
- ▶ After an empty survey, each of the other two blocks has oil probability $1/2$.
- ▶ Better information must be worth its acquisition cost.

Expected Utility: Main Takeaways

- ▶ Choose actions using expected utility, accounting for preferences over outcomes.
- ▶ Selecting among noisy utility estimates can introduce optimism even when each estimate is unbiased.
- ▶ Value information by the expected improvement from actions chosen after observing it.
- ▶ Compare information value with its cost, using compatible units and the current evidence.

Thank You

Any questions?



References

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